

INDIAN INSTITUTE OF TECHNOLOGY BOMBAY
ELECTRICAL ENGINEERING DEPARTMENT

EE-Midsem

Monday Sep. 18, 2023
 Time: 17:15-18:00

MS-101 Makerspace
 Autumn Semester 2023

Marks: 30
 (To be re-scaled)

Roll Number:

Name:

Division:

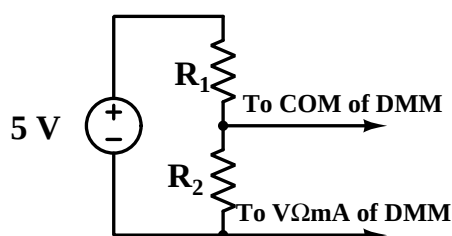
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Signature:

This question paper has 8 questions over 4 pages.

Answer the following in the space provided with the questions.

Q-1 In the circuit shown below, two resistors R_1 and R_2 are connected to the 5 V DC output of the Keithley DC Power Supply.



$R_1 = R_2 = 2 \text{ M}\Omega$. (Note: $1 \text{ M}\Omega = 10^6 \Omega$).

The voltage across R_2 was measured using a DMM, by putting it in the DC Voltage range.

The DMM in the chosen voltage range can show two decimal places and has an internal resistance of $10 \text{ M}\Omega$.

What reading will you see on the DMM?

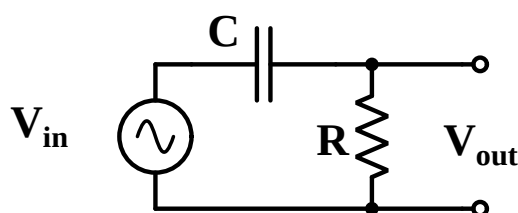
Answer:

V

– [2]

Q-2 Circuit diagram of an RC high-pass filter is shown below in the figure.

$R = 2 \text{ k}\Omega$ and $C = 0.01 \mu\text{F}$.



The input-output expression for this filter can be written as:

$$\frac{V_{out}}{V_{in}} = \frac{R}{R + (1/j\omega C)}$$

A) What is the cut-off frequency f_c in Hz of this RC filter?

Answer:

$f_c =$ Hz

– [1]

B) Find the magnitude of V_{out}/V_{in}

Answer:

i) at f_c

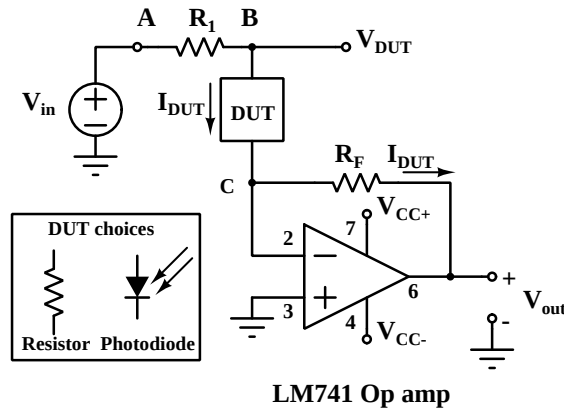
V_{out}/V_{in} at $f_c =$

ii) at 2 kHz

V_{out}/V_{in} at 2 kHz =

– [1 + 2]

Q-3 Circuit diagram of the op amp I-to-V converter used in Expt 2 is shown below.



The device under test (DUT) is connected between terminals B and C.

Assume the op amp is ideal.

$V_{cc+} = +12\text{ V}$, $V_{cc-} = -12\text{ V}$;

$R_1 = 1.5\text{ k}\Omega$, $R_F = 2\text{ k}\Omega$.

Also assume that the maximum and minimum V_{out} levels are V_{cc+} and V_{cc-} respectively.

Calculation steps must be shown with your answer. No marks will be awarded without proper steps.

- a) A $3\text{ k}\Omega$ resistor is connected as the DUT. If $V_{in} = +6\text{ V}$, what will be the values of V_{DUT} and V_{out} ?

Calculation Steps:

Answer:

$V_{DUT} =$	V
$V_{out} =$	V

– [1 + 1]

- b) A photodiode (similar to the one used in the lab) is connected as the DUT with its anode at terminal B and cathode at terminal C. Assume that light from a mobile flashlight is made to fall on the photodiode and the corresponding photocurrent is $200\text{ }\mu\text{A}$. V_{in} is a triangular wave of 200 Hz with +ve and –ve peaks of $+8\text{ V}$ and -8 V respectively.

- i) Sketch the I-V characteristic of this photodiode as seen on the DSO when it is put in the X-Y mode with V_{DUT} as X and I_{DUT} as Y and with the Y channel inverted. Indicate appropriate voltage and current values.

Sketch:

– [2]

- ii) Based on the I-V characteristic of the photodiode in the reverse biased region, comment with justification whether the photodiode is working as a voltage source or as a current source.

Comment with justification:

– [2]

- Q-4 a)** A sine wave with peak to peak amplitude of approximately 200 mV is superimposed on a DC value of about 10 V.

Choose the correct coupling mode (AC or DC) for accurately measuring:

- i) the average value of this waveform

AC / DC

and

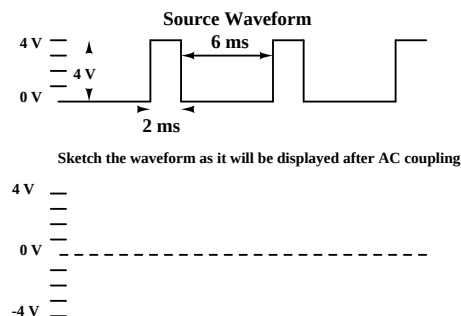
- ii) the peak to peak value of the sine wave.

AC / DC

– [2]

- b)** Consider a source waveform as shown in the figure below. This wave is displayed on an oscilloscope using AC coupling.

Sketch how the waveform will appear on the oscilloscope. (The waveform after AC coupling should be drawn with an identical time scale as used for the source waveform and should show the voltage levels at the bottom and top of the waveform in text).



– [3]

- Q-5** Using 3 bits, we can represent unsigned numbers in the range (0-7). We want to design a combinational digital circuit with a 3 bit input whose output is ‘1’ if the number represented by the input bits is a prime number and ‘0’ otherwise. (Numbers 2, 3, 5 and 7 are prime numbers in this range).

- a)** Using the table on the right (where A, B and C are inputs with A as the most significant bit), fill in the truth table entries for the function F whose output should be ‘1’ whenever the input number is a prime number.

A	B	C	F
0	0	0	
0	0	1	
0	1	0	
0	1	1	
1	0	0	
1	0	1	
1	1	0	
1	1	1	

- b)** Express the function F in canonical form as a sum of products.

Answer: $F =$

– [3]

- Q-6** A function is specified in canonical sum of products form as

$$F = \overline{A}\overline{B}\overline{C}D + \overline{A}\overline{B}C\overline{D} + \overline{A}B\overline{C}\overline{D} + \overline{A}B\overline{C}D + A\overline{B}\overline{C}D + A\overline{B}C\overline{D} + AB\overline{C}\overline{D} + ABC\overline{D}$$

Fill in entries for this function in the Karnaugh map given on the right.

Find the minimal expression for this function using the Karnaugh minimisation procedure.

CD ↓ AB →	00	01	11	10
00				
01				
11				
10				

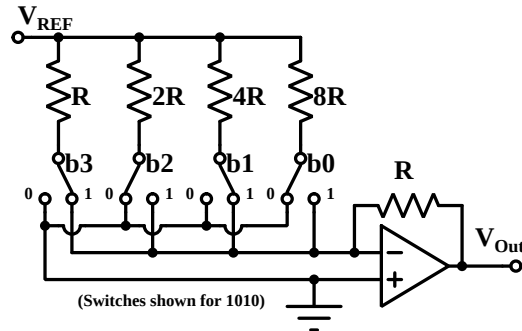
Answer: $F =$

– [5]

Q-7 An n bit A-D converter divides a given range of analog values in 2^n equal intervals and represents any analog value in the range by the interval number enclosing it. This involves a possible inaccuracy of half the size of the interval in the worst case. This is called quantization error.

What is the least number of bits that one should use for digital representation of analog voltages in the range 0 to 5V such that the worst case quantization error is less than 5 mV. – [2]

Q-8 Consider a 4 bit D-A converter using binary weighted resistors as shown in the figure below.



Each switch is controlled by a bit of the digital word being converted to analog and steers the current through the resistor (to which it is connected) to ground or to virtual ground depending on the bit being '0' or '1'.

For $V_{REF} = 4V$, Fill in the output voltage V_{out} for the given digital values in the table below:

Digital Word $b_3b_2b_1b_0$	Output voltage V_{out} in V
1100	
1010	
0111	

– [3]

_____ Paper Ends _____

Rough Work